

```
mysql> create database Views_MySQL;  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> show databases;
```

Database
bookflow_db
db2
employeedb1
information_schema
mysql
performance_schema
practice1
practice2
practice3
stored_procedures
studentdb1
sys
views_mysql

```
13 rows in set (0.00 sec)
```

```
mysql> use views_MySQL;
```

```
Database changed
```

```
mysql> CREATE TABLE IF NOT EXISTS departments (  
->     dept_id INT PRIMARY KEY,  
->     dept_name VARCHAR(50)  
-> );
```

```
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE IF NOT EXISTS employees (  
->     emp_id INT PRIMARY KEY,  
->     emp_name VARCHAR(50),  
->     dept_id INT,  
->     salary DECIMAL(10,2)  
-> );
```

```
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> INSERT IGNORE INTO departments VALUES
-> (1, 'CSE'),
-> (2, 'ECE'),
-> (3, 'EEE');
```

```
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT IGNORE INTO employees VALUES
-> (101, 'Rahul', 1, 50000),
-> (102, 'Priya', 2, 65000),
-> (103, 'Anil', 1, 55000),
-> (104, 'Sneha', 3, 70000),
-> (105, 'Kiran', 2, 48000);
```

```
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> show tables;
```

```
+-----+
| Tables_in_views_mysql |
+-----+
| departments           |
| employees              |
+-----+
```

```
2 rows in set (0.01 sec)
```

```
mysql> select * from departments;
```

```
+-----+-----+
| dept_id | dept_name |
+-----+-----+
| 1       | CSE      |
| 2       | ECE      |
+-----+-----+
```

```
mysql> select * from employees;
```

emp_id	emp_name	dept_id	salary
101	Rahul	1	50000.00
102	Priya	2	65000.00
103	Anil	1	55000.00
104	Sneha	3	70000.00
105	Kiran	2	48000.00

```
5 rows in set (0.00 sec)
```

```
mysql> DROP VIEW IF EXISTS EmployeeView;
```

```
Query OK, 0 rows affected, 1 warning (0.00 sec)
```

```
mysql>
```

```
mysql> CREATE VIEW EmployeeView AS
```

```
-> SELECT *
```

```
-> FROM employees;
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
```

```
mysql> SELECT * FROM EmployeeView;
```

emp_id	emp_name	dept_id	salary
101	Rahul	1	50000.00
102	Priya	2	65000.00
103	Anil	1	55000.00
104	Sneha	3	70000.00
105	Kiran	2	48000.00

```
5 rows in set (0.00 sec)
```

```
mysql> DROP VIEW IF EXISTS EmployeeBasicView;
```

```
Query OK, 0 rows affected, 1 warning (0.00 sec)
```

```
mysql>
```

```
mysql> DROP VIEW IF EXISTS EmployeeBasicView;  
Query OK, 0 rows affected, 1 warning (0.00 sec)
```

```
mysql>  
mysql> CREATE VIEW EmployeeBasicView AS  
-> SELECT emp_id, emp_name, salary  
-> FROM employees;  
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>  
mysql> SELECT * FROM EmployeeBasicView;
```

emp_id	emp_name	salary
101	Rahul	50000.00
102	Priya	65000.00
103	Anil	55000.00
104	Sneha	70000.00
105	Kiran	48000.00

```
5 rows in set (0.00 sec)
```

```
mysql> DROP VIEW IF EXISTS HighSalaryEmployees;  
Query OK, 0 rows affected, 1 warning (0.00 sec)
```

```
mysql>  
mysql> CREATE VIEW HighSalaryEmployees AS  
-> SELECT emp_id, emp_name, salary  
-> FROM employees  
-> WHERE salary > 55000;  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM HighSalaryEmployees;
```

emp_id	emp_name	salary
102	Priya	65000.00
104	Sneha	70000.00

```
2 rows in set (0.00 sec)
```

```
mysql> DROP VIEW IF EXISTS EmployeeDepartmentView;  
Query OK, 0 rows affected, 1 warning (0.00 sec)
```

```
mysql>
```

```
mysql> CREATE VIEW EmployeeDepartmentView AS
```

```
-> SELECT
```

```
->     e.emp_id,
```

```
->     e.emp_name,
```

```
->     d.dept_name,
```

```
->     e.salary
```

```
-> FROM employees e
```

```
-> JOIN departments d
```

```
-> ON e.dept_id = d.dept_id;
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
```

```
mysql> SELECT * FROM EmployeeDepartmentView;
```

emp_id	emp_name	dept_name	salary
101	Rahul	CSE	50000.00
102	Priya	ECE	65000.00
103	Anil	CSE	55000.00
104	Sneha	EEE	70000.00
105	Kiran	ECE	48000.00

```
5 rows in set (0.00 sec)
```

```
mysql> DROP VIEW IF EXISTS CSEEmployees;
Query OK, 0 rows affected, 1 warning (0.00 sec)
```

```
mysql>
mysql> CREATE VIEW CSEEmployees AS
-> SELECT
->     e.emp_id,
->     e.emp_name,
->     d.dept_name,
->     e.salary
-> FROM employees e
-> JOIN departments d
-> ON e.dept_id = d.dept_id
-> WHERE d.dept_name = 'CSE';
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> SELECT * FROM CSEEmployees;
+-----+-----+-----+-----+
| emp_id | emp_name | dept_name | salary |
+-----+-----+-----+-----+
|    101 | Rahul    | CSE       | 50000.00 |
|    103 | Anil     | CSE       | 55000.00 |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> DROP VIEW IF EXISTS DepartmentSalaryView;
Query OK, 0 rows affected, 1 warning (0.00 sec)
```

```
mysql>
mysql> CREATE VIEW DepartmentSalaryView AS
-> SELECT
->     d.dept_name,
->     COUNT(e.emp_id) AS EmployeeCount,
->     AVG(e.salary) AS AverageSalary
-> FROM employees e
-> JOIN departments d
-> ON e.dept_id = d.dept_id
-> GROUP BY d.dept_id, d.dept_name;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> DROP VIEW IF EXISTS HighAverageDepartments;  
Query OK, 0 rows affected, 1 warning (0.00 sec)
```

```
mysql>
```

```
mysql> CREATE VIEW HighAverageDepartments AS  
-> SELECT  
->     d.dept_name,  
->     AVG(e.salary) AS AverageSalary  
-> FROM employees e  
-> JOIN departments d  
-> ON e.dept_id = d.dept_id  
-> GROUP BY d.dept_id, d.dept_name  
-> HAVING AVG(e.salary) > 55000;  
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
```

```
mysql> SELECT * FROM HighAverageDepartments;
```

dept_name	AverageSalary
ECE	56500.000000
EEE	70000.000000

```
2 rows in set (0.00 sec)
```

```
mysql> DROP VIEW IF EXISTS AnnualSalaryView;  
Query OK, 0 rows affected, 1 warning (0.01 sec)
```

```
mysql>
```

```
mysql> CREATE VIEW AnnualSalaryView AS  
-> SELECT  
->     emp_id,  
->     emp_name,  
->     salary AS MonthlySalary,  
->     salary * 12 AS AnnualSalary  
-> FROM employees;
```

```
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM AnnualSalaryView;
```

emp_id	emp_name	MonthlySalary	AnnualSalary
101	Rahul	50000.00	600000.00
102	Priya	65000.00	780000.00
103	Anil	55000.00	660000.00
104	Sneha	70000.00	840000.00
105	Kiran	48000.00	576000.00

```
5 rows in set (0.00 sec)
```

```
mysql> DROP VIEW IF EXISTS EmployeeSalaryView;  
Query OK, 0 rows affected, 1 warning (0.00 sec)
```

```
mysql>
```

```
mysql> CREATE VIEW EmployeeSalaryView AS  
-> SELECT emp_id, emp_name, salary  
-> FROM employees;
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
```

```
mysql> UPDATE EmployeeSalaryView  
-> SET salary = 60000  
-> WHERE emp_id = 101;
```

```
Query OK, 1 row affected (0.00 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql>
```

```
mysql> SELECT * FROM EmployeeSalaryView;
```

emp_id	emp_name	salary
101	Rahul	60000.00
102	Priya	65000.00
103	Anil	55000.00
104	Sneha	70000.00
105	Kiran	48000.00

```
5 rows in set (0.00 sec)
```



```
mysql> SELECT * FROM employees WHERE emp_id = 101;
```

emp_id	emp_name	dept_id	salary
101	Rahul	1	60000.00

```
1 row in set (0.00 sec)
```

```
mysql> CREATE OR REPLACE VIEW EmployeeBasicView AS  
-> SELECT emp_name, salary  
-> FROM employees;
```

```
Query OK, 0 rows affected (0.01 sec)
```

```
mysql>
```

```
mysql> SELECT * FROM EmployeeBasicView;
```

emp_name	salary
Rahul	60000.00
Priya	65000.00
Anil	55000.00
Sneha	70000.00
Kiran	48000.00

```
5 rows in set (0.00 sec)
```

```
mysql> SHOW CREATE VIEW EmployeeDepartmentView;
```

```

+-----+
+-----+-----+-----+
| View          | Create View                                     | character_set_client | collation_connection |
+-----+-----+-----+-----+
+-----+-----+-----+-----+
| employee_departmentview | CREATE ALGORITHM=UNDEFINED DEFINER=`root`@`localhost` SQL SECURITY DEFINER VIEW `employee_departmentview` AS
S select `e`.`emp_id` AS `emp_id`,`e`.`emp_name` AS `emp_name`,`d`.`dept_name` AS `dept_name`,`e`.`salary` AS `salary` from (`employee
es` `e` join `departments` `d` on((`e`.`dept_id` = `d`.`dept_id`))) | cp850 | cp850_general_ci
+-----+-----+-----+-----+
+-----+-----+-----+-----+

```

Tables_in_views_mysql	Table_type
annualsalaryview	VIEW
cseemployees	VIEW
departmentsalaryview	VIEW
employeebasicview	VIEW
employeedepartmentview	VIEW
employeesalaryview	VIEW
employeeview	VIEW
highaveragedepartments	VIEW
highsalaryemployees	VIEW

```
mysql> DROP VIEW IF EXISTS SalaryRangeView;
Query OK, 0 rows affected, 1 warning (0.00 sec)
```

```
mysql> CREATE VIEW EmpBasic AS
-> SELECT emp_id, emp_name, salary
-> FROM employees;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE VIEW HighEarners AS
-> SELECT emp_id, emp_name, salary
-> FROM employees
-> WHERE salary > 60000;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE VIEW EmpDeptSalary AS
-> SELECT e.emp_name, d.dept_name, e.salary
-> FROM employees e
-> JOIN departments d ON e.dept_id = d.dept_id;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE VIEW ECEmployees AS
-> SELECT e.emp_id, e.emp_name, d.dept_name, e.salary
-> FROM employees e
-> JOIN departments d ON e.dept_id = d.dept_id
-> WHERE d.dept_name = 'ECE';
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> select * from EmpBasic;
```

emp_id	emp_name	salary
101	Rahul	60000.00
102	Priya	65000.00
103	Anil	55000.00
104	Sneha	70000.00
105	Kiran	48000.00

```
5 rows in set (0.00 sec)
```

mysql> select * from EmpBasic;

emp_id	emp_name	salary
101	Rahul	60000.00
102	Priya	65000.00
103	Anil	55000.00
104	Sneha	70000.00
105	Kiran	48000.00

5 rows in set (0.00 sec)

mysql> select * from HighEarners;

emp_id	emp_name	salary
102	Priya	65000.00
104	Sneha	70000.00

2 rows in set (0.00 sec)

mysql> select * from EmpDeptSalary;

emp_name	dept_name	salary
Rahul	CSE	60000.00
Priya	ECE	65000.00
Anil	CSE	55000.00
Sneha	EEE	70000.00
Kiran	ECE	48000.00

5 rows in set (0.00 sec)

mysql> select * from ECEEmployees;

emp_id	emp_name	dept_name	salary
102	Priya	ECE	65000.00

```
mysql> select * from ECEEmployees;
```

emp_id	emp_name	dept_name	salary
102	Priya	ECE	65000.00
105	Kiran	ECE	48000.00

```
2 rows in set (0.00 sec)
```

```
mysql> CREATE VIEW DeptEmployeeCount AS
```

```
-> SELECT d.dept_name, COUNT(e.emp_id) AS EmployeeCount
```

```
-> FROM employees e
```

```
-> JOIN departments d ON e.dept_id = d.dept_id
```

```
-> GROUP BY d.dept_name;
```

```
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> select * from DeptEmployeeCount;
```

dept_name	EmployeeCount
CSE	2
ECE	2
EEE	1

```
3 rows in set (0.00 sec)
```

```
mysql> CREATE VIEW DeptMaxSalary AS
```

```
-> SELECT d.dept_name, MAX(e.salary) AS MaxSalary
```

```
-> FROM employees e
```

```
-> JOIN departments d ON e.dept_id = d.dept_id
```

```
-> GROUP BY d.dept_name;
```

```
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> select * from DeptMaxSalary;
```

dept_name	MaxSalary
CSE	60000.00
ECE	65000.00

```
mysql> CREATE VIEW DeptHighAvg AS
-> SELECT d.dept_name, AVG(e.salary) AS AvgSalary
-> FROM employees e
-> JOIN departments d ON e.dept_id = d.dept_id
-> GROUP BY d.dept_name
-> HAVING AVG(e.salary) > 50000;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> select * from DeptHighAvg;
+-----+-----+
| dept_name | AvgSalary |
+-----+-----+
| CSE       | 57500.000000 |
| ECE       | 56500.000000 |
| EEE       | 70000.000000 |
+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> CREATE VIEW EmpAnnualSalary AS
-> SELECT emp_name, salary, salary * 12 AS AnnualSalary
-> FROM employees;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> select * from EmpAnnualSalary;
+-----+-----+-----+
| emp_name | salary | AnnualSalary |
+-----+-----+-----+
| Rahul    | 60000.00 | 720000.00 |
| Priya    | 65000.00 | 780000.00 |
| Anil     | 55000.00 | 660000.00 |
| Sneha    | 70000.00 | 840000.00 |
| Kiran    | 48000.00 | 576000.00 |
+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> CREATE VIEW EmpSalaryView AS
-> SELECT emp_id, emp_name, salary
-> FROM employees;
Query OK, 0 rows affected (0.00 sec)
```

```

mysql>
mysql> UPDATE EmpSalaryView
  -> SET salary = 62000
  -> WHERE emp_id = 102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SELECT * FROM employees WHERE emp_id = 102;
+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | salary |
+-----+-----+-----+-----+
|    102 | Priya    |        2 | 62000.00 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> CREATE VIEW EmpSalaryView AS
  -> SELECT emp_id, emp_name, salary
  -> FROM employees;
ERROR 1050 (42S01): Table 'EmpSalaryView' already exists
mysql>
mysql> UPDATE EmpSalaryView
  -> SET salary = 62000
  -> WHERE emp_id = 102;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0

mysql>
mysql> SELECT * FROM employees WHERE emp_id = 102;
+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | salary |
+-----+-----+-----+-----+
|    102 | Priya    |        2 | 62000.00 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

```

```
mysql> SHOW CREATE VIEW EmpDeptSalary;
+-----+-----+
| View          | Create View                                     |
+-----+-----+
|               | character_set_client | collation_connection |
+-----+-----+
| empdeptsalary | CREATE ALGORITHM=UNDEFINED DEFINER='root'@'localhost' SQL SECURITY DEFINER VIEW `empdeptsalary` AS select `e`.`emp_name` AS `emp_name`,`d`.`dept_name` AS `dept_name`,`e`.`salary` AS `salary` from (`employees` `e` join `departments` `d` on((`e`.`dept_id` = `d`.`dept_id`))) | cp850 | cp850_general_ci |
+-----+-----+
1 row in set (0.00 sec)
```



```
mysql> SHOW FULL TABLES
      -> WHERE Table_type = 'VIEW';
```

Tables_in_views_mysql	Table_type
annualsalaryview	VIEW
cseemployees	VIEW
departmentsalaryview	VIEW
deptemployeecount	VIEW
depthhighavg	VIEW
deptmaxsalary	VIEW
eceemployees	VIEW
empannualsalary	VIEW
empbasic	VIEW
empdeptsalary	VIEW
employeebasicview	VIEW
employeedepartmentview	VIEW
employeesalaryview	VIEW
employeeview	VIEW
empsalaryview	VIEW
highaveragedepartments	VIEW
highearners	VIEW
highsalaryemployees	VIEW

18 rows in set (0.00 sec)

```
mysql> DROP VIEW IF EXISTS DeptMaxSalary;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql>
```

```
mysql> SHOW FULL TABLES
```

```
    -> WHERE Table_type = 'VIEW';
```

Tables_in_views_mysql	Table_type
annualsalaryview	VIEW
cseemployees	VIEW
departmentsalaryview	VIEW
deptemployeecount	VIEW
depthhighavg	VIEW
eceemployees	VIEW
empannualsalary	VIEW
empbasic	VIEW
empdeptsalary	VIEW
employeebasicview	VIEW
employeeedepartmentview	VIEW
employeesalaryview	VIEW
employeeview	VIEW
empsalaryview	VIEW
highaveragedepartments	VIEW
highearners	VIEW
highsalaryemployees	VIEW

```
17 rows in set (0.00 sec)
```

```
mysql> |
```